

Problem 1: Bie field and chafe fix

(1)

Given

$$\Sigma_{\text{bief}} \text{fix}(S) = \frac{1}{2} [I + U_1 X + (1-2p) U_2 Y + (1-2p) U_3 Z] \quad \left. \vphantom{\Sigma_{\text{bief}} \text{fix}(S)} \right\} \text{Prove this}$$

$$\Sigma_{\text{chafe}} \text{fix}(S) = \frac{1}{2} [I + (1-2p) U_1 X + (1-2p) U_2 Y + U_3 Z]$$

$$S = \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z]$$

Solution

The operators elements for Bie fix

$$E_0 = \sqrt{1-p} I$$

$$E_1 = \sqrt{p} X$$

$$(\delta p)^+ = \delta p$$

$$\Sigma(S) = E_0 S E_0^+ + E_1 S E_1^+$$

$$= \sqrt{1-p} I S (\sqrt{1-p} I)^+ + \sqrt{p} X S (\sqrt{p} X)^+$$

$$= (\sqrt{1-p})^2 I S I + (p) X S X$$

$$= (1-p) S + p X S X$$

$$= (1-p) \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z] + p X \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z] X$$

$$= \frac{1}{2} [(1-p) (I + U_1 X + U_2 Y + U_3 Z) + p X (I + U_1 X + U_2 Y + U_3 Z) X]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - p - p U_1 X - p U_2 Y - p U_3 Z + p X X + p X U_1 X X + p X U_2 Y X + p X U_3 Z X]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - p - p U_1 X - p U_2 Y - p U_3 Z + p \underbrace{X X}_{=I} + p U_1 \underbrace{X X X}_{=X} + p U_2 \underbrace{X Y X}_{=Y} + p U_3 \underbrace{X Z X}_{=Z}]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - p - p U_1 X - p U_2 Y - p U_3 Z + p U_1 X - p U_2 Y - p U_3 Z]$$

(2)

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - \underbrace{p+p}_{=0} - \underbrace{pU_1 X + pU_1 X}_{=0} - pU_2 Y - pU_2 Y - pU_3 Z - pU_3 Z]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - 2pU_2 Y - 2pU_3 Z]$$

$$= \frac{1}{2} [I + U_1 X + (\cancel{U_2 Y} - 2pU_2 Y) + (\cancel{U_3 Z} - 2pU_3 Z)]$$

$$= \frac{1}{2} [I + U_1 X + (1-2p)U_2 Y + (1-2p)U_3 Z]$$

$$\underline{\underline{\rho(\mathcal{B}) = \frac{1}{2} [I + U_1 X + (1-2p)U_2 Y + (1-2p)U_3 Z]}}$$

The operation element for phase-flip channel

$$E_0 = \sqrt{1-p} I$$

$$E_1 = \sqrt{p} Z$$

$$\rho(\mathcal{B}) = E_0 \mathcal{B} E_0^\dagger + E_1 \mathcal{B} E_1^\dagger$$

$$= \sqrt{1-p} I \mathcal{B} (\sqrt{1-p} I)^\dagger + \sqrt{p} Z \mathcal{B} (\sqrt{p} Z)^\dagger$$

$$= (\sqrt{1-p})^2 I \mathcal{B} I + (\sqrt{p})^2 Z \mathcal{B} Z$$

$$= (1-p) I \mathcal{B} I + p Z \mathcal{B} Z$$

$$= (1-p) \mathcal{B} + p Z \mathcal{B} Z$$

$$= (1-p) \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z] + p Z \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z] Z$$

$$= \frac{1}{2} [(1-p) [I + U_1 X + U_2 Y + U_3 Z] + p Z [I + U_1 X + U_2 Y + U_3 Z] Z]$$

$$= \frac{1}{2} [(1-p) (I + U_1 X + U_2 Y + U_3 Z) + p Z (I + U_1 X + U_2 Y + U_3 Z) Z]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - \rho - \rho U_1 X - \rho U_2 Y - \rho U_3 Z + \rho Z Z + \rho U_1 Z X Z + \rho U_2 Z Y Z + \rho U_3 Z Z Z]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - \rho - \rho U_1 X - \rho U_2 Y - \rho U_3 Z + \rho - \rho U_1 X - \rho U_2 Y + \rho U_3 Z]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z - \underbrace{\rho + \rho}_{=0} - \rho U_1 X - \rho U_1 X - \rho U_2 Y - \rho U_2 Y - \underbrace{\rho U_3 Z + \rho U_3 Z}_{=0}]$$

$$= \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z] - 2\rho U_1 X - 2\rho U_2 Y$$

$$= \frac{1}{2} [I + (U_1 X - 2\rho U_1 X) + (U_2 Y - 2\rho U_2 Y) + U_3 Z]$$

$$Z(\rho) = \frac{1}{2} [I + (\lambda - 2\rho) U_1 X + (\lambda - 2\rho) U_2 Y + U_3 Z]$$

2. Comparing

Solution

The transformed $S' = Z(\rho)$, $S' = \frac{1}{2} [I + U' \cdot \sigma]$

For BPT Flip Channel scaling

(Comparing) $Z_{\text{BPT flip}}(\rho) = \frac{1}{2} [I + U_1 X + (\lambda - 2\rho) U_2 Y + (\lambda - 2\rho) U_3 Z]$ to

$$S' = \frac{1}{2} [I + U'_1 X + U'_2 Y + U'_3 Z]$$

The Bloch Vector

$$\left. \begin{array}{ll} x\text{-axis,} & U_1 \rightarrow \rho U_1 \\ y\text{-axis,} & U_2 \rightarrow \rho(\lambda - 2\rho) U_2 \\ z\text{-axis,} & U_3 \rightarrow (\lambda - 2\rho) U_3 \end{array} \right\}$$

For Phase Flip Channel

Comparing, $Z_{\text{phase flip}}(\rho) = \frac{1}{2} [I + (\lambda - 2\rho) U_1 X + (\lambda - 2\rho) U_2 Y + U_3 Z]$

$$\text{to } S = \frac{1}{2} [I + U_1 X + U_2 Y + U_3 Z]$$

The Bloch Vector

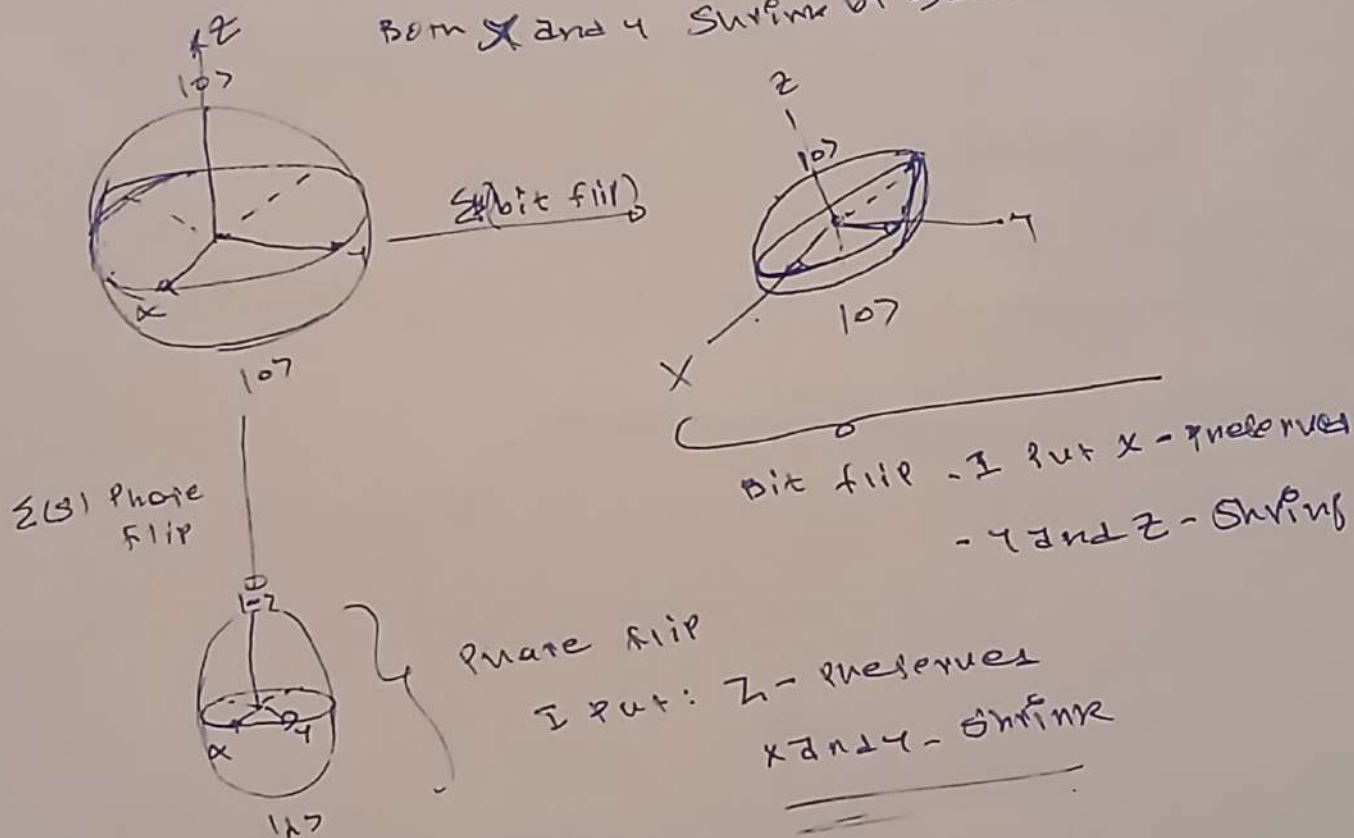
$$\begin{array}{lcl} \text{x-axis} & U_1 & \rightarrow (1-2p)U_1 \\ \text{y-axis} & U_2 & \rightarrow (1-2p)U_2 \\ \text{z-axis} & U_3 & \rightarrow U_3 \end{array}$$

Draw the corresponding scaling of the Bloch sphere

To draw the effect of two channel

Bit flip channel: x-axis unchanged
Both y and z shrink by scale of $(1-2p)$

Phase flip channel: z-axis unchanged
Both x and y shrink by scale of $(1-2p)$



3. What is stationary state i.e. the state that are invariant under the mapping for the two channels?

Solution

$$S(1) = \frac{1}{2} [I + U_1 X + (1-2p)U_2 Y + (1-2p)U_3 Z]$$

$$S(1) = \frac{1}{2} [I + (1-2p)U_1 X + (1-2p)U_2 Y + (1-2p)U_3 Z]$$

For the Bit flip

$$z(x) = \frac{1}{2} [I + U_1 X + (\lambda - 2P) U_2 Y + (\lambda - 2P) U_3 Z]$$

Stationary requires

$$V_1 = V_1, \quad (\lambda - 2P) V_2 = V_2, \quad (\lambda - 2P) V_3 = V_3$$

$$\lambda = 2$$

$$\lambda - 2P \neq 1$$

$$\lambda - 2P \neq \lambda$$

$$\underbrace{\hspace{10em}}$$

$$V_2 = 0, \quad V_3 = 0$$

$$\underbrace{\hspace{10em}}$$

$$= \underline{\underline{V_2 = V_3 = 0}}$$

$$z(x) = \frac{1}{2} [I + U_1 X + \underbrace{(\lambda - 2P) 0 \cdot Y + (\lambda - 2P) 0 \cdot Z}_{= 0}]$$

$$= \underline{\underline{\frac{1}{2} [I + U_1 X]}} \quad , \quad |U_1| \leq 1$$

For Bit Phase flip

$$z(x) = \frac{1}{2} [I + (\lambda - 2P) U_2 X + (\lambda - 2P) U_2 Y + \cancel{(\lambda - 2P) U_3 Z}]$$

$$= \frac{1}{2} [I + (\lambda - 2P) U_2 X + (\lambda - 2P) U_2 Y + U_3 Z]$$

Stationary requires

$$V_1 = (\lambda - 2P) V_1, \quad V_2 = (\lambda - 2P) V_2, \quad V_3 = V_3$$

$$\lambda \neq \lambda - 2P$$

$$\lambda \neq \lambda - 2P$$

$$\lambda = \lambda$$

$$\underbrace{\hspace{10em}}$$

$$V_1 = 0$$

$$V_2 = 0$$

$$\underbrace{\hspace{10em}}$$

$$\underline{\underline{V_1 = V_2 = 0}}$$

$$\hat{Z}(\theta) = \frac{1}{2} [I + (\lambda - 2\mu) \cdot 0 \cdot X + (\lambda - 2\mu) \cdot 0 \cdot Y + U_Z Z]$$

$$= \frac{1}{2} [I + 0 + 0 + U_Z Z]$$

$$= \frac{1}{2} [I + U_Z Z], \text{ For, } \underline{|U_Z| \leq 1}$$

Where U_Z to U_X + U_Y shows

$$\hat{Z}_{\text{bit flip}}(\theta) = \frac{1}{2} [I + U_X X]$$

$$\hat{Z}_{\text{phase flip}}(\theta) = \frac{1}{2} [I + U_Z Z]$$

Therefore, bit flip state along the X-axis of Bloch Sphere
and stationary state was eigenstate of Pauli X which is
 $|+\rangle$ and $|-\rangle$ state

for phase flip

↳ The State along Z-axis on Bloch Sphere

↳ It was state of Pauli ~~X~~ Z which means eigenstate

of $|0\rangle$ and $|1\rangle$

Problem: 2 determining, channel

2.

Solution

$$Z(s) = p \cdot \frac{1}{2} + (s-p) Z$$

$$= p \cdot \frac{1}{4} [s + xsx + 1sy + zsz] + (s-p) Z$$

$$= \frac{p}{4} [s + xsx + 1sy + zsz] + (s-p) Z$$

$$= \frac{ps}{4} + \frac{p}{4} [xsx + 1sy + zsz] + (s-p) Z$$

$$= \frac{ps}{4} + (s-p) Z + \frac{p}{4} [xsx + 1sy + zsz]$$

$$= \frac{ps}{4} + \frac{4(s-p)Z}{4} + \frac{p}{4} [xsx + 1sy + zsz]$$

$$= \frac{ps + 4s - 4pZ}{4} + \frac{p}{4} [xsx + 1sy + zsz]$$

$$= \frac{4s - 3pZ}{4} + \frac{p}{4} [xsx + 1sy + zsz]$$

$$= \frac{(4 - 3p)Z}{4} + \frac{p}{4} [xsx + 1sy + zsz]$$

$$= \left(\frac{4 - 3p}{4} \right) Z + \frac{p}{4} [xsx + 1sy + zsz]$$

Klaus Operator:

$$E_0 = \sqrt{\frac{4 - 3p}{4}} I, \quad E_i = \frac{\sqrt{p}}{2} \sigma_i \text{ for } i = 1, 2, 3$$

2) Solution

$$Z(s) = p \cdot \frac{1}{2} + (s-p) Z$$

$$Z(s) = \frac{1}{2}$$

$$p \cdot \frac{1}{2} + (s-p) Z = \frac{1}{2}$$

$$p \cdot \frac{1}{2} + (s-p) Z - \frac{1}{2} = 0$$

$$p \cdot \frac{I}{2} - \frac{I}{2} + (p-r)S = 0$$

$$\frac{I}{2} [p-r] + [p-r]S = 0$$

$$\frac{I}{2} [p-r] - [p-r]S = 0$$

$$\frac{[p-r] [\frac{I}{2} - S]}{[\frac{I}{2} - S]} = 0$$

$$[p-r] = 0$$

$$p-r = 0$$

$$\underline{\underline{p = r}}$$

Problem 3: Amplitude and phase damping

1) Solution

$$Z(s) = E_0 S E_0^+ + E_r S E_r^+$$

$$\text{Hence, } E_0 = 107401 + \sqrt{1-r} 117411 = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-r} \end{bmatrix}$$

$$E_0^+ = E_0$$

$$E_r = \sqrt{r} 107411 = \begin{bmatrix} 0 & \sqrt{r} \\ 0 & 0 \end{bmatrix}$$

$$E_r^+ = \sqrt{r} 117401 = \begin{bmatrix} 0 & 0 \\ \sqrt{r} & 0 \end{bmatrix}$$

$$S = \begin{bmatrix} S_{00} & S_{0r} \\ S_{r0} & S_{rr} \end{bmatrix}$$

$$Z(s) = E_0 S E_0^+ + E_r S E_r^+$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-r} \end{bmatrix} \begin{bmatrix} S_{00} & S_{0r} \\ S_{r0} & S_{rr} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-r} \end{bmatrix} + \begin{bmatrix} 0 & \sqrt{r} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} S_{00} & S_{0r} \\ S_{r0} & S_{rr} \end{bmatrix} \sqrt{r}$$

$$\begin{aligned}
 Z(s) &= \begin{bmatrix} s_{00} & s_{0r} \\ \sqrt{1-r} s_{h0} & \sqrt{1-r} s_{hr} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-r} \end{bmatrix} + \begin{bmatrix} \sqrt{r} s_{h0} & \sqrt{r} s_{hr} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ \sqrt{r} & 0 \end{bmatrix} \\
 &= \begin{bmatrix} s_{00} & \sqrt{1-r} s_{0r} \\ \sqrt{1-r} s_{h0} & (1-r) s_{hr} \end{bmatrix} + \begin{bmatrix} (\sqrt{r})^2 s_{hr} & 0 \\ 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} s_{00} & \sqrt{1-r} s_{0r} \\ \sqrt{1-r} s_{h0} & (1-r) s_{hr} \end{bmatrix} + \begin{bmatrix} r s_{hr} & 0 \\ 0 & 0 \end{bmatrix} \\
 &= \begin{pmatrix} s_{00} + r s_{hr} & \sqrt{1-r} s_{0r} \\ \sqrt{1-r} s_{h0} & (1-r) s_{hr} \end{pmatrix}
 \end{aligned}$$

2) Solution

$$Z(s) = Z_1(Z_2(s))$$

$$Z_2(s) = \begin{bmatrix} s_{00} + r_2 s_{hr} & \sqrt{1-r_2} s_{0r} \\ \sqrt{1-r_2} s_{h0} & (1-r_2) s_{hr} \end{bmatrix}$$

$Z_1(Z_2(s))$ Let say, $A = Z_2(s)$ for simplify the expression

$$\begin{aligned}
 Z_1(A) &= \begin{bmatrix} A_{00} + r_1 A_{hr} & \sqrt{1-r_1} A_{0r} \\ \sqrt{1-r_1} A_{h0} & (1-r_1) A_{hr} \end{bmatrix} \quad \text{where, } A_{00} = s_{00} + r_2 s_{hr} \\
 & \quad A_{0r} = \sqrt{1-r_2} s_{0r} \\
 & \quad A_{h0} = \sqrt{1-r_2} s_{h0} \\
 & \quad A_{hr} = (1-r_2) s_{hr}
 \end{aligned}$$

$$\begin{aligned}
 Z(s) &= Z_1(Z_2(s)) \\
 &= \begin{bmatrix} s_{00} + r_2 s_{hr} + r_1 (1-r_2) s_{hr} & \sqrt{1-r_1} \sqrt{1-r_2} s_{0r} \\ \sqrt{1-r_1} \sqrt{1-r_2} s_{h0} & (1-r_1)(1-r_2) s_{hr} \end{bmatrix} \\
 &= \begin{bmatrix} s_{00} + (r_2 + r_1(1-r_2)) s_{hr} & \sqrt{1-r_1} \sqrt{1-r_2} s_{0r} \\ \sqrt{1-r_1} \sqrt{1-r_2} s_{h0} & (1-r_1)(1-r_2) s_{hr} \end{bmatrix}
 \end{aligned}$$

$$Z(s) = \begin{bmatrix} s_{00} + (r_2 + r_1(s-r_2)) s_{21} & \sqrt{s-r_1} \sqrt{s-r_2} s_{02} \\ \sqrt{s-r_1} \sqrt{s-r_2} s_{10} & (s-r_1)(s-r_2) s_{11} \end{bmatrix}$$

parameter $Z(s) = Z_1(s) Z_2(s)$

$$1 - s_{00} + r_1 s_{11} = s_{00} + (r_2 + r_1(s-r_2)) s_{11}$$

$$2 - \sqrt{s-r_1} s_{10} = \sqrt{s-r_1} \sqrt{s-r_2} s_{10}$$

$$3 - \sqrt{s-r_1} \sqrt{s-r_2} s_{02} = \sqrt{s-r_1} s_{02}$$

$$4 - (s-r_1) s_{11} = (s-r_1)(s-r_2) s_{11}$$

But I want take one to show the parameter

let's take - number - (1) to show the same for all,

$$s_{00} + r_1 s_{11} = s_{00} + (r_2 + r_1(s-r_2)) s_{11}$$

$$s_{00} - s_{00} + r_1 s_{11} = (r_2 + r_1(s-r_2)) s_{11}$$

$$0 + r_1 s_{11} = [r_2 + r_1(s-r_2)] s_{11}$$

$$\frac{r_1 s_{11}}{s_{11}} = [r_2 + r_1(s-r_2)] \frac{s_{11}}{s_{11}}$$

$$r = r_2 + r_1 - r_1 r_2$$

$$r = \underline{\underline{r_2 + r_1 - r_1 r_2}}$$

$$\underline{\underline{r = r_2 + r_1 - r_1 r_2}}$$

3) pole damping

$$E_0 = \frac{1}{2} \omega_{L01} + \sqrt{\lambda - \lambda} \frac{1}{2} \omega_{L01} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}, \quad \text{where } E_0 = E_0^+$$

$$E_1 = \sqrt{\lambda} \frac{1}{2} \omega_{L01} = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$E_1 = E_1^+$$

Solution

$$Z(s) = E_0 S E_0^T + E_1 S E_1^T$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix} \begin{pmatrix} g_{00} & g_{01} \\ g_{10} & g_{11} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix} \begin{pmatrix} g_{00} & g_{01} \\ g_{10} & g_{11} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$$

$$= \begin{pmatrix} g_{00} & g_{01} \\ \sqrt{1-\lambda} g_{10} & \sqrt{1-\lambda} g_{11} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \sqrt{\lambda} g_{10} & \sqrt{\lambda} g_{11} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$$

$$= \begin{pmatrix} g_{00} & \sqrt{1-\lambda} g_{01} \\ \sqrt{1-\lambda} g_{10} & \sqrt{1-\lambda} \sqrt{1-\lambda} g_{11} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \sqrt{\lambda} g_{11} \end{pmatrix}$$

$$= \begin{pmatrix} g_{00} & \sqrt{1-\lambda} g_{01} \\ \sqrt{1-\lambda} g_{10} & (1-\lambda) g_{11} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \lambda g_{11} \end{pmatrix}$$

$$= \begin{bmatrix} g_{00} + 0 & \sqrt{1-\lambda} g_{01} + 0 \\ 0 + \sqrt{1-\lambda} g_{10} & (1-\lambda) g_{11} + \lambda g_{11} \end{bmatrix} = \begin{bmatrix} g_{00} & \sqrt{1-\lambda} g_{01} \\ \sqrt{1-\lambda} g_{10} & (1-\lambda + \lambda) g_{11} \end{bmatrix}$$

$$\underline{\underline{Z(s) = \begin{pmatrix} g_{00} & \sqrt{1-\lambda} g_{01} \\ \sqrt{1-\lambda} g_{10} & g_{11} \end{pmatrix}}}$$

1) Solution

The phase flip channel are

$$K_0 = \sqrt{1-p} I = \begin{pmatrix} \sqrt{1-p} & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}$$

$$K_1 = \sqrt{p} Z = \begin{pmatrix} \sqrt{p} & 0 \\ 0 & -\sqrt{p} \end{pmatrix}$$

The phase damping channel

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}$$

Unitary Relating Kraus Operators

$$\begin{pmatrix} K_0 \\ K_1 \end{pmatrix} = U \begin{pmatrix} E_0 \\ E_1 \end{pmatrix} \Rightarrow \begin{pmatrix} K_0 \\ K_1 \end{pmatrix} = \begin{pmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{pmatrix} \begin{pmatrix} E_0 \\ E_1 \end{pmatrix}$$

$$\begin{pmatrix} K_0 \\ K_1 \end{pmatrix} = \begin{pmatrix} U_{00} E_0 + U_{01} E_1 \\ U_{10} E_0 + U_{11} E_1 \end{pmatrix}$$

$$K_0 = U_{00} E_0 + U_{01} E_1$$

$$\begin{pmatrix} \overline{1-\rho} & 0 \\ 0 & \overline{1-\rho} \end{pmatrix} = U_{00} \begin{pmatrix} 1 & 0 \\ 0 & \overline{1-\lambda} \end{pmatrix} + U_{01} \begin{pmatrix} 0 & 0 \\ 0 & \overline{1-\lambda} \end{pmatrix}$$

$$\begin{pmatrix} \overline{1-\rho} & 0 \\ 0 & \overline{1-\rho} \end{pmatrix} = \begin{pmatrix} U_{00} & 0 \\ 0 & U_{00} \overline{1-\lambda} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & U_{01} \overline{1-\lambda} \end{pmatrix}$$

$$\cancel{U_{00}} \begin{pmatrix} \overline{1-\rho} & 0 \\ 0 & \overline{1-\rho} \end{pmatrix} = \begin{pmatrix} U_{00} & 0 \\ 0 & U_{00} \overline{1-\lambda} + U_{01} \overline{1-\lambda} \end{pmatrix}$$

$$\underline{\underline{U_{00} = \overline{1-\rho} \quad \checkmark}}$$

$$\underline{\underline{U_{00} \overline{1-\lambda} + U_{01} \overline{1-\lambda} = \overline{1-\rho}}}$$

$$= \overline{1-\rho}$$

$$\overline{1-\rho} \overline{1-\lambda} + U_{01} \overline{1-\lambda} = \overline{1-\rho}$$

where as, $\overline{1-\lambda} = 1-2\rho$, which is we get from $\sum \text{phase flip}(S) = \sum \text{phase damping}(S)$

$\sum \text{phase flip}(S) = \sum \text{phase damping}(S)$, we get

$$\underline{\underline{\overline{1-\lambda} = 1-2\rho}}$$

$$\overline{a-p} (a-2p) + W_{02} \overline{a} = \overline{a-p}$$

$$W_{02} \overline{a} = \overline{a-p} - [\overline{a-p} (a-2p)]$$

$$W_{02} \overline{a} = \overline{a-p} - [\overline{a-p} (a-2p)]$$

where

$$\overline{a-a} = a-2p$$

$$a-a = (a-2p)^2$$

$$a-a = (a-2p)(a-2p)$$

$$a-a = a-a-4p+4p^2$$

$$\cancel{a-a} = -4p+4p^2$$

$$-a = -4p+4p^2$$

$$a = 4p-4p^2$$

$$\overline{a} = \sqrt{4p(a-p)}$$

$$\overline{a} = 2\sqrt{p(a-p)}$$

$$\overline{a} = 2\sqrt{p(a-p)}$$

$$W_{02} \overline{a} = \overline{a-p} - [\overline{a-p} (a-2p)]$$

$$W_{02} \cdot 2\sqrt{p(a-p)} = \overline{a-p} - \overline{a-p} (a-2p)$$

$$W_{02} \cdot 2\sqrt{p} \overline{a-p} = \overline{a-p} [a-(a-2p)]$$

$$W_{02} \cdot 2\sqrt{p} \overline{a-p} = \overline{a-p} [a-a+2p]$$

$$\frac{W_{02} \cdot 2\sqrt{p} \overline{a-p}}{2\sqrt{p} \overline{a-p}} = \frac{\overline{a-p} (2p)}{2\sqrt{p} \overline{a-p}}$$

$$W_{02} = \frac{p}{\sqrt{p}}$$

$$W_{02} = \frac{p}{\sqrt{p}} \times \frac{\sqrt{p}}{\sqrt{p}}$$

$$\underline{\underline{W_{02} = \sqrt{p}}}$$

When go to the second equation?

$$k_t = W_{00} E_0 + W_{01} E_1$$

$$\begin{pmatrix} \sqrt{p} & 0 \\ 0 & \sqrt{p} \end{pmatrix} = W_{00} \begin{bmatrix} a & 0 \\ 0 & \overline{a-a} \end{bmatrix} + W_{01} \begin{bmatrix} 0 & 0 \\ 0 & \overline{a} \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{p} & 0 \\ 0 & \sqrt{p} \end{bmatrix} = \begin{bmatrix} W_{00} & 0 \\ 0 & W_{00} \overline{a-a} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & W_{01} \overline{a} \end{bmatrix}$$

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$$\begin{bmatrix} \sqrt{p} & 0 \\ 0 & -\sqrt{p} \end{bmatrix} = \begin{bmatrix} W_{20} & 0 \\ 0 & W_{20}\sqrt{2-p} + W_{22}\sqrt{p} \end{bmatrix}$$

$$\underline{W_{20} = \sqrt{p}} \quad \checkmark$$

$$W_{20}\sqrt{2-p} + W_{22}\sqrt{p} = -\sqrt{p}$$

$$\sqrt{p}\sqrt{2-p} + W_{22}\sqrt{p} = -\sqrt{p}$$

$$\sqrt{p}(2-2p) + W_{22}(2\sqrt{p}\sqrt{2-p}) = -\sqrt{p}$$

$$\frac{\sqrt{p}}{\sqrt{p}} [(2-2p) + W_{22}2\sqrt{2-p}] = \frac{-\sqrt{p}}{\sqrt{p}}$$

$$(2-2p) + W_{22}2\sqrt{2-p} = -1$$

$$W_{22}2\sqrt{2-p} = -2 - (2-2p)$$

$$W_{22}2\sqrt{2-p} = -2 - (2-2p)$$

$$W_{22}2\sqrt{2-p} = -2 - 2 + 2p$$

$$W_{22}2\sqrt{2-p} = -2 + 2p$$

$$\frac{W_{22}2\sqrt{2-p}}{2\sqrt{2-p}} = \frac{-2(2-p)}{2\sqrt{2-p}}$$

$$W_{22} = -\frac{(2-p)}{\sqrt{2-p}}$$

$$W_{22} = -\frac{(2-p) \times \sqrt{2-p}}{\sqrt{2-p} \sqrt{2-p}}$$

$$W_{22} = \frac{-(2-p) \times \sqrt{2-p}}{(2-p)}$$

$$W_{22} = -(\sqrt{2-p})$$

→ Finally

$$W_{00} = \sqrt{2-p}$$

$$W_{02} = \sqrt{p}$$

$$W_{20} = \sqrt{p}$$

$$W_{22} = -\sqrt{2-p}$$

$$W = \begin{bmatrix} W_{00} & W_{02} \\ W_{20} & W_{22} \end{bmatrix}$$

$$W = \begin{bmatrix} \sqrt{2-p} & \sqrt{p} \\ \sqrt{p} & -\sqrt{2-p} \end{bmatrix}$$